

Exploratory Data Analysis on Zomato Restaurant Dataset

1. Introduction

Online food delivery platforms generate large volumes of data related to restaurants, customer preferences, pricing, and services. Analyzing this data helps in understanding market trends and consumer behavior. This project focuses on performing Exploratory Data Analysis (EDA) on the Zomato restaurant dataset to extract meaningful insights using data analysis and visualization techniques.

2. Objective of the Project

The main objectives of this project are: - To understand the structure and characteristics of the Zomato dataset - To clean and preprocess the data for analysis - To analyze restaurant trends based on ratings, cost, cuisines, and online delivery - To visualize patterns and relationships between important features - To derive business-oriented insights from the data

3. Dataset Description

The Zomato dataset contains information about restaurants listed on the Zomato platform. The dataset includes the following key attributes: - Restaurant Name - City / Location - Cuisines - Average Cost for Two - Aggregate Rating - Online Delivery availability - Table Booking availability

The dataset consists of several thousand records representing restaurants across different cities.

4. Tools and Technologies Used

- Programming Language: Python
 - Libraries:
 - Pandas (data manipulation)
 - NumPy (numerical operations)
 - Matplotlib (data visualization)
 - Seaborn (statistical visualization)
 - Environment: Jupyter Notebook
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5. Methodology

5.1 Data Loading

The dataset was loaded into the Jupyter Notebook using the Pandas library. Initial inspection was done using functions such as `head()`, `info()`, and `describe()` to understand data types and distributions.

5.2 Data Cleaning and Preprocessing

- Checked for missing values and handled them appropriately
- Converted rating values from string format to numeric values
- Removed or treated invalid and missing ratings
- Ensured consistency in categorical variables such as online delivery

This step was crucial to ensure reliable analysis results.

6. Exploratory Data Analysis

6.1 Distribution of Restaurants

Analysis was performed to understand how restaurants are distributed across cities and locations. This helps identify regions with higher restaurant density.

6.2 Ratings Analysis

- Studied the distribution of restaurant ratings
- Identified the most common rating ranges
- Observed that most restaurants fall under mid to high rating categories

6.3 Cost for Two Analysis

- Analyzed the distribution of average cost for two people
- Identified common pricing ranges
- Detected the presence of high-cost outliers

6.4 Online Delivery Impact

- Compared ratings of restaurants with and without online delivery
- Observed that restaurants offering online delivery tend to have slightly higher ratings

6.5 Cuisine Analysis

- Identified the most popular cuisines
- Compared cuisines based on average ratings
- Observed that certain cuisines consistently receive higher customer ratings

7. Data Visualization

Various plots were used to visualize insights, including: - Bar charts for categorical comparisons - Count plots for frequency analysis - Distribution plots for ratings and cost - Comparative plots for online delivery vs ratings

These visualizations made it easier to interpret trends and patterns in the data.

8. Key Insights

- Majority of restaurants fall in the mid-price range
 - Online delivery availability positively influences ratings
 - Certain cuisines are more popular and better rated than others
 - Restaurant distribution is uneven across cities
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9. Conclusion

This Exploratory Data Analysis provided valuable insights into restaurant trends and customer preferences on the Zomato platform. The findings can help restaurant owners and food delivery platforms make informed decisions related to pricing, service offerings, and market expansion.

10. Future Scope

- Apply machine learning models to predict restaurant ratings
 - Perform sentiment analysis on customer reviews
 - Create interactive dashboards using tools like Power BI or Tableau
 - Conduct city-specific deep-dive analysis
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11. References

- Zomato Dataset
- Pandas Documentation
- Seaborn Documentation